



Simultaneous Masked and Unmasked Decoding with Speculative Decoding Masking for Fast ASR without Accuracy Loss

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Abstract

In this paper, we introduce two methods, Simultaneous Masked and Unmasked Decoding (SMUD) and speculative decoding masking, into Partially autoregressive (PAR) decoding. These methods achieve the same recognition accuracy as Autoregressive (AR) decoding while maintaining higher computational efficiency than AR in Automatic Speech Recognition (ASR). SMUD and speculative decoding masking can accurately identify hypotheses where decoder score computation can be omitted. By omitting these computations, they achieve faster processing while obtaining the same search results as AR decoding.

In TED-LIUM2 evaluations, SMUD with speculative decoding masking achieved a WER of 7.3% and an RTF of 0.41, as compared to AR's WER of 7.3% and RTF of 0.59, showcasing the method's ability to maintain the same high accuracy as AR while enhancing computational efficiency.

Index Terms: Automatic speech recognition (ASR), partially autoregressive (PAR) decoding, speculative decoding, beam search, CTC-decoder joint decoding

1. Introduction

In recent years, high-accuracy ASR using Transformer-based encoder-decoder models which employ Autoregressive (AR) decoding, has become feasible [1, 2, 3, 4, 5]. While AR decoding is highly accurate, it suffers from the issue of extended processing times caused by iterative decoder processing. As a consequence, Non-autoregressive (NAR) decoding methods [6, 7, 8, 9, 10, 11, 12] have been proposed to address this issue. Although NAR methods do offer faster processing, they have not yet achieved accuracy comparable to that of AR methods [13]. In many ASR applications, high recognition accuracy is required, leading to the continued use of highly accurate AR in many ASR engines, including Whisper [3]. Given this background, there is a demand for faster search methods than AR that do not compromise its accuracy.

To bridge this gap, Partially Autoregressive (PAR) decoding [14] has been proposed to integrate the benefits of AR's accuracy and NAR's speed. This method generates preliminary recognition results using CTC greedy search and employs AR decoding only in regions with low CTC posterior probabilities. While PAR decoding is faster than AR decoding and more accurate than NAR decoding, it still suffers from poor accuracy as compared to AR decoding.

In this paper, we have identified the primary causes for the poor accuracy of PAR when compared to AR: early search termination, inability to use CTC prefix scores, and low mask prediction accuracy. To resolve these issues and achieve fast decoding while maintaining the accuracy of AR, we introduce

two methods into PAR: Simultaneous Masked and Unmasked Decoding (SMUD) and speculative decoding masking.

SMUD has two key characteristics. The first is that while searching for hypotheses within the masked region, it simultaneously calculates decoder scores for both the hypotheses within the masked region and those considered to have exited it to outside the masked region. By comparing these scores, SMUD can accurately determine whether the search within the mask can be terminated. The second characteristic is that, regardless of whether a hypothesis is inside or outside the mask, it always calculates CTC prefix scores, enabling highly accurate CTC-decoder joint decoding. These characteristics help maintain the high recognition accuracy of AR. Furthermore, speculative decoding masking enhances mask prediction accuracy by utilizing decoder outputs in addition to CTC greedy search results for predicting the mask. Specifically, the entire preliminary recognition result from the CTC greedy search is fed into the decoder at once, and positions in the input tokens are further masked where the output tokens differ from them. This approach continues to enable fast mask generation, as it processes the entire recognition result in one batch. By combining these methods, we achieve fast inference speeds without accuracy degradation as compared to AR decoding.

2. Preliminary

2.1. AR with CTC-decoder joint decoding

Autoregressive (AR) decoding predicts tokens one by one, using the previously predicted token sequence as decoder input to predict the next token iteratively, as shown in Figure 1 (a). This method involves using beam search to predict the next token for the top B hypotheses, and it is widely recognized as one of the most accurate search methods in ASR, including Whisper [3].

Further, [15, 16] have proposed a method that combines the CTC prefix score with the neural network decoder score in AR decoding. The CTC prefix score is the log value of the CTC prefix probability, which is defined as the cumulative probability of all token sequences that have the current hypothesis as their prefix. By using both scores together, higher recognition accuracy can be achieved, further enhancing the advantages of AR. For example, OWSM [4, 5] utilizes this joint-decoding technique.

However, to obtain the recognition result of L tokens using AR, it is necessary to iteratively input the intermediate recognition results into the decoder L times, resulting in slower processing. This slowdown occurs not only due to increased computational complexity but also because each time an input is fed, CPU cache misses occur, requiring the neural network's weight parameters to be read from memory.

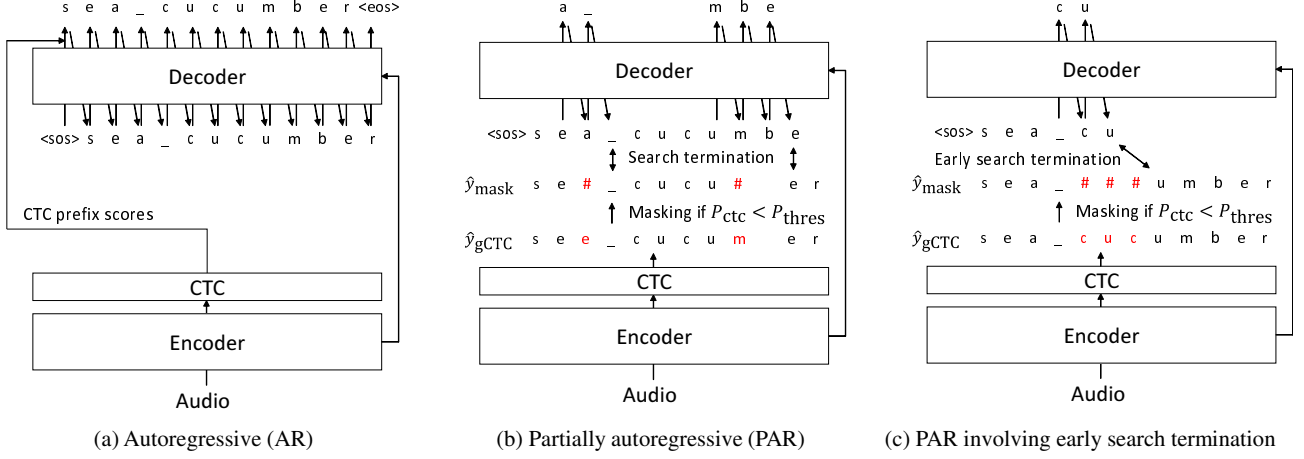


Figure 1: (a) Autoregressive (AR) with CTC-decoder joint decoding, high-accuracy but slow due to iterative decoder processing; (b) Partially autoregressive (PAR), faster with fewer iterative decoder processing only for masked regions; (c) Example of early search termination in PAR, leading to deletion errors.

2.2. Partially autoregressive decoding

Partially autoregressive (PAR) decoding [14], as illustrated in Figure 1 (b), initially obtains a preliminary recognition result ($\hat{y}_{gCTC}$) using fast CTC greedy search (gCTC) in an NAR manner. Afterward, tokens with CTC probabilities lower than P_{thres} are masked with mask tokens (#), resulting in the masked token sequence $\hat{y}_{mask}$, where only these masked tokens are re-predicted using high accuracy AR decoding. By partially employing AR decoding, re-predicted segments maintain the high recognition accuracy of AR decoding, without the large degradation seen in NAR methods [13]. Moreover, computational cost is reduced as compared to full AR decoding because the computationally expensive iterative decoder processing is only applied to the masked regions and skipped for the unmasked regions.

However, despite these advantages, PAR has three issues that lead to degraded accuracy compared to AR:

1. **Early search termination:** The termination criteria of beam search in masked regions are based on the appearance of tokens following the mask (post-mask tokens). Beam search may terminate early if the same tokens as the post-mask tokens appear within the mask regions. This issue will be discussed in detail later, on the basis of an example.
2. **Inability to use CTC prefix scores:** PAR cannot utilize CTC prefix scores, which were used in AR to enhance recognition accuracy, because CTC prefix scores cannot be computed solely within the mask. They need to be calculated sequentially from the beginning of the input, including regions outside the mask. In PAR, which limits its search to within the mask, it is not possible to compute CTC prefix scores.
3. **Low mask prediction accuracy:** Since CTC does not achieve as high recognition accuracy as the decoder, the mask prediction accuracy using CTC probabilities is insufficient. If the incorrectly predicted tokens by gCTC are output with a higher probability than P_{thres} and are not masked, these incorrect tokens remain in $\hat{y}_{mask}$, leading to recognition errors.

To address the issue of early search termination, consider an example shown in Figure 1 (c) where the correct transcription, using character-based tokens, for a given utterance is *s e a _ c u c u m b e r*.

Suppose that the tokens *c u c* are masked due to their low CTC probabilities, and as a result, the masked token sequence $\hat{y}_{mask}$ becomes *s e a _ # # # u m b e r*. If we define a post-mask token as a token that immediately follows the masked region, in this example, the post-mask token would be *u*. Thus, the appearance of *u* would trigger the hypothesis search termination criteria during beam search. When performing beam search within the masked region, *c* might be selected as the first token in the mask region, followed by *u* as the next token. Since the post-mask token *u* appears, the search terminates early, and the recognition result for the mask region, which would correctly be *c u c*, instead becomes *c*. As a result, the overall recognition output is *s e a _ c u m b e r*. Thus, early search termination can lead to deletion errors.

3. Simultaneous masked and unmasked decoding

In this section, we propose Simultaneous Masked and Unmasked Decoding (SMUD), a method based on PAR. From the block diagram of PAR in Figure 1 (b), SMUD enhances it by modifying the search termination criteria and utilizing CTC prefix scores. This approach addresses two accuracy degradation issues of PAR, specifically early search termination and the inability to use CTC prefix scores, and achieves recognition accuracy comparable to AR.

The following explains the algorithm of SMUD using Figure 2, demonstrating how it realizes precise search termination for masked regions. As in the previous section, we use the example where the correct transcription for a given utterance is *s e a _ c u c u m b e r*, while the masked token sequence $\hat{y}_{mask}$ is *s e a _ # # # u m b e r*.

- Step 1 The process in this step is the same as in PAR. For the first unmasked segment, *<sos> s e a _*, the decoder score calculation is omitted. For the first masked segment, AR decoding with beam search is started. The hypotheses searched within this masked segment are referred to as the mask hypotheses (H_{in}).
- Step 2 During beam search, suppose a hypothesis becomes *<sos> s e a _ c u*, and the post-mask token appears

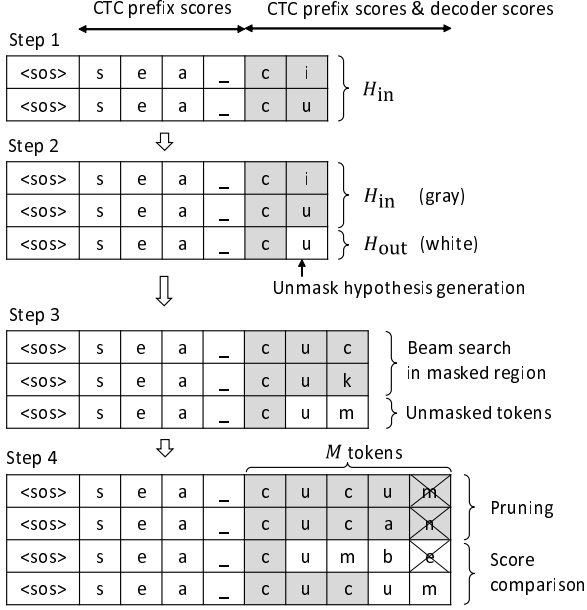


Figure 2: Simultaneous masked and unmasked decoding, which avoids early search termination and enables CTC-decoder joint decoding (with a beam size of 2).

(in this example, u). Unlike PAR, SMUD considers two possibilities: the token u might truly be the post-mask token appearing immediately after the mask, or the same token as the post-mask token might have appeared in the middle of the mask. To address the possibility that it is a token appearing in the middle of the mask, the mask hypothesis is retained as is. On the other hand, to consider the possibility that it is a post-mask token appearing after the mask, we generate an unmask hypothesis (H_{out}), which is considered to have exited the mask while having the exact same token sequence.

Step 3 Continue the beam search for H_{in} . Simultaneously, for H_{out} , extend the hypothesis by assuming that the next unmasked tokens appear according to $\hat{y}_{mask}$, such as u m b e r in the example. In this step, calculate the decoder scores for both H_{in} and H_{out} .

Step 4 Steps 2 and 3 are repeated. During this repetition, a new unmask hypothesis is generated for s e a _ c u c u. The beam search for H_{in} is conducted for M tokens ($M = 5$ in this example), pruning any mask hypothesis that does not reach a post-mask token at this point. In this example, the two mask hypotheses in H_{in} , s e a _ c u c u m and s e a _ c u c a n are pruned. Finally, if there are multiple unmask hypotheses in H_{out} , the scores are compared, retaining only the hypothesis with the highest score. In this example, the hypothesis with the correct recognition result, s e a _ c u c u m, has a higher score than s e a _ c u m b e, and thus the former hypothesis is retained.

The SMUD algorithm compares the scores of multiple hypotheses where the post-mask token appears at different positions and retains the most likely hypothesis. As a result, early search termination is avoided, enabling SMUD to maintain recognition accuracy equivalent to that of AR decoding.

Furthermore, by computing the scores as described below,

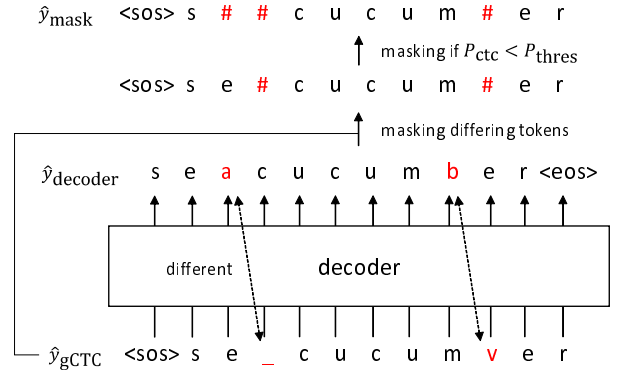


Figure 3: Speculative decoding masking, which provides a more accurate mask prediction by utilizing the decoder.

SMUD is able to employ CTC-decoder joint decoding. For the hypothesis h_i at position i , the score S_i to be added to the cumulative score is calculated as follows:

$$S_i = \begin{cases} \alpha S_{ctc,i} & \text{if } \forall h_i, h_i \in H_{out}, \\ \alpha S_{ctc,i} + (1 - \alpha) S_{decoder,i} & \text{otherwise.} \end{cases} \quad (1)$$

In essence, if all hypotheses at position i are unmask hypotheses, H_{out} , which corresponds to a column where all boxes are white in Figure 2, the decoder score computation is skipped, and only the CTC prefix score $S_{ctc,i}$ is calculated. Otherwise, if there is even one mask hypothesis, H_{in} , both the CTC prefix score $S_{ctc,i}$ and the decoder score $S_{decoder,i}$ are computed, enabling CTC-decoder joint decoding. This approach enables the use of highly accurate CTC-decoder joint decoding for the re-prediction of masked regions, thereby maintaining the high accuracy of AR decoding. Additionally, since the decoder score is not computed for regions where all hypotheses are H_{out} , consistent with PAR, the search process is faster than that of AR decoding. In the case of CPU decoding, the computational cost of CTC prefix scores is relatively small compared to that of computing decoder scores [16], allowing SMUD to maintain its speed even without skipping the calculation of CTC prefix scores for unmasked regions.

4. Speculative decoding masking

We propose speculative decoding masking (SDmask), which provides a more accurate mask prediction than using only gCTC. Since gCTC has lower recognition accuracy, it often outputs incorrect tokens with high probability, resulting in incorrect unmasked tokens. Therefore, we also utilize a more accurate decoder for mask prediction as well. However, to maintain speed, we calculate the decoder scores using a NAR approach.

Below is a detailed explanation of the process. As shown in Figure 3, speculative decoding masking involves inputting the entire token sequence $\hat{y}_{gCTC}$ obtained from gCTC into the decoder at once, predicting the subsequent tokens across the entire sequence, and obtaining $\hat{y}_{decoder}$. Similar to speculative decoding, a technique known for speeding up large language models (LLM) inference [17], this method can predict the entire sequence of L tokens in a single input pass, making it more cache-efficient and keeping the masking process fast.

Next, $\hat{y}_{gCTC}$ and $\hat{y}_{decoder}$ are compared, and the differing tokens are replaced with mask tokens. Additionally, as in the con-

Table 1: CER or WER(%). Values in each column represent CER or WER for dev/test sets. For LS-100, values represent WER for dev_clean/dev_other/test_clean/test_other. SMUD with SDmask achieves recognition accuracy comparable to AR.

Method	CTC weight	SDmask	AISHELL-1	JSUT	LS-100	TED-LIUM2
AR	0.3		4.2 / 4.5	11.6 / 13.3	6.3 / 16.9 / 6.5 / 17.1	7.3 / 7.2
PAR [14]	0.0		4.4 / 4.7	11.9 / 13.3	6.4 / 17.1 / 6.6 / 17.4	7.6 / 7.3
SMUD	0.3		4.2 / 4.5	11.6 / 13.2	6.4 / 17.1 / 6.6 / 17.4	7.4 / 7.2
SMUD	0.3	✓	4.2 / 4.5	11.6 / 13.2	6.3 / 16.9 / 6.5 / 17.2	7.3 / 7.2

Table 2: Real-time factors (RTFs). SMUD with SDmask is faster than AR.

Method	CTC weight	SDmask	AISHELL-1	JSUT	LS-100	TED-LIUM2
AR	0.3		0.67	0.67	0.47	0.59
PAR [14]	0.0		0.57	0.47	0.33	0.32
SMUD	0.3		0.60	0.53	0.39	0.40
SMUD	0.3	✓	0.61	0.55	0.40	0.41

ventional method, parts of the sequence where the gCTC probability is lower than P_{thres} are also masked. By combining these two types of mask predictions, more accurate mask predictions are achieved.

5. Experiments

5.1. Experimental setup

We used four E-Branchformer models [2, 18] that were used in the experiments conducted in [14]. The E-Branchformer models were pre-trained on the following datasets and are publicly available on the HuggingFace hub: AISHELL-1¹, JSUT², LibriSpeech-100h (LS-100)³, TED-LIUM2⁴. All models were trained and evaluated using the ESPnet toolkit [19].

Common settings for AR decoding, PAR decoding, and SMUD include a beam size of 5, and no external language model, with inference conducted on a CPU (Xeon Gold 6230R). The CTC weight was 0.3 for AR and SMUD, 0.0 for PAR. In common with both PAR and SMUD, P_{thres} was set to 0.95, and the maximum search length M was set to 5. Additionally, further experiments were conducted by combining SMUD with speculative decoding masking.

For this study, we used several datasets for evaluation, similarly to what is described in [14]. Specifically, we used LibriSpeech [20] and TED-LIUM2 [21] as English datasets, JSUT [22] as a Japanese dataset, and AISHELL-1 [23] as a Mandarin dataset.

We followed the evaluation metrics described in [14] as well. We evaluated accuracy using the word error rate (WER) or character error rate (CER) depending on the dataset. Specifically, we used CER for the AISHELL-1 and JSUT evaluation sets, and WER for English evaluation sets. The real-time factor (RTF) was used to measure inference speed. For the RTF, we took the mean of all evaluation sets.

5.2. Results

Experimental results for CER and WER are presented in Table 1, and RTF values are shown in Table 2.

It can be observed that PAR achieves smaller RTFs than does AR, but at the cost of increased recognition error due to the

issues identified in this paper. For example, on TED-LIUM2, the RTF decreases from 0.59 to 0.32, while the WER on the dev set increases from 7.3% to 7.6%.

On the other hand, SMUD employing CTC-decoder joint decoding and precise search termination exhibits only slight degradation in recognition error compared to AR. For instance, the WER on the TED-LIUM2 dev set is 7.4%. Although the RTF for SMUD is 0.40, which is higher than PAR’s 0.32 due to the calculation of CTC prefix scores, it still achieves a speedup over AR’s 0.59.

Moreover, combining SMUD with speculative decoding masks further reduces accuracy degradation, achieving recognition accuracy nearly identical to that of AR. For example, the WER on the TED-LIUM2 dev set is 7.3%. The additional processing time introduced by speculative decoding masks is minimal, resulting in only a slight increase in RTF. TED-LIUM2, for instance, shows an RTF of 0.41, which is significantly faster than AR’s 0.59.

From the above results, it can be concluded that combining SMUD with speculative decoding masking achieves slightly lower speed than PAR but with improved recognition accuracy over PAR. More importantly, it delivers recognition accuracy comparable to AR while being faster than AR.

6. Conclusion

In this study, we have introduced Simultaneous Masked and Unmasked Decoding (SMUD) into fast PAR decoding to maintain the high accuracy of AR decoding. SMUD enables precise search termination and CTC-decoder joint decoding to address the accuracy degradation in PAR while preserving its speed advantage over AR. Additionally, we have proposed the speculative decoding masking to enhance mask prediction and further address the accuracy degradation of PAR.

Our experimental results confirm the effectiveness of SMUD and speculative decoding masking. Unlike PAR decoding, which exhibits noticeable degradation in recognition accuracy, SMUD maintains high recognition accuracy with significantly better computational efficiency than AR. Specifically, in TED-LIUM2 evaluations, SMUD with speculative decoding masking achieved a WER of 7.3% and an RTF of 0.41, as compared to AR’s WER of 7.3% and RTF of 0.59. This demonstrates that SMUD, combined with speculative decoding masking, offers a faster method than traditional AR decoding with nearly no degradation in accuracy.

¹https://huggingface.co/pyf98/aishell_e_branchformer

²https://huggingface.co/pyf98/jsut_e_branchformer

³https://huggingface.co/pyf98/librispeech_100_e_branchformer

⁴https://huggingface.co/pyf98/tedlium2_e_branchformer

7. References

- [1] A. Gulati, J. Qin, C.-C. Chiu, N. Parmar, Y. Zhang, J. Yu, W. Han, S. Wang, Z. Zhang, Y. Wu, and R. Pang, “Conformer: Convolution-augmented Transformer for Speech Recognition,” in *Proc. Interspeech 2020*, 2020, pp. 5036–5040.
- [2] K. Kim, F. Wu, Y. Peng, J. Pan, P. Sridhar, K. J. Han, and S. Watanabe, “E-branchformer: Branchformer with enhanced merging for speech recognition,” in *2022 IEEE Spoken Language Technology Workshop (SLT)*. IEEE, 2023, pp. 84–91.
- [3] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, “Robust speech recognition via large-scale weak supervision,” in *International conference on machine learning*. PMLR, 2023, pp. 28 492–28 518.
- [4] Y. Peng, J. Tian, B. Yan, D. Berrebbi, X. Chang, X. Li, J. Shi, S. Arora, W. Chen, R. Sharma *et al.*, “Reproducing whisper-style training using an open-source toolkit and publicly available data,” in *2023 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*. IEEE, 2023, pp. 1–8.
- [5] Y. Peng, J. Tian, W. Chen, S. Arora, B. Yan, Y. Sudo, M. Shakeel, K. Choi, J. Shi, X. Chang, J. weon Jung, and S. Watanabe, “Owsm v3.1: Better and faster open whisper-style speech models based on e-branchformer,” in *Interspeech 2024*, 2024, pp. 352–356.
- [6] Y. Higuchi, S. Watanabe, N. Chen, T. Ogawa, and T. Kobayashi, “Mask CTC: Non-Autoregressive End-to-End ASR with CTC and Mask Predict,” in *Proc. Interspeech 2020*, 2020, pp. 3655–3659.
- [7] Y. Higuchi, H. Inaguma, S. Watanabe, T. Ogawa, and T. Kobayashi, “Improved mask-ctc for non-autoregressive end-to-end asr,” in *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2021, pp. 8363–8367.
- [8] J. Lee and S. Watanabe, “Intermediate loss regularization for ctc-based speech recognition,” in *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2021, pp. 6224–6228.
- [9] J. Nozaki and T. Komatsu, “Relaxing the conditional independence assumption of ctc-based asr by conditioning on intermediate predictions,” in *Interspeech 2021*, 2021, pp. 3735–3739.
- [10] R. Fan, W. Chu, P. Chang, and J. Xiao, “Cass-nat: Ctc alignment-based single step non-autoregressive transformer for speech recognition,” in *ICASSP 2021 - 2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021, pp. 5889–5893.
- [11] Y. Bai, J. Yi, J. Tao, Z. Tian, Z. Wen, and S. Zhang, “Fast end-to-end speech recognition via non-autoregressive models and cross-modal knowledge transferring from bert,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 29, pp. 1897–1911, 2021.
- [12] Z. Gao, S. Zhang, I. McLoughlin, and Z. Yan, “Paraformer: Fast and accurate parallel transformer for non-autoregressive end-to-end speech recognition,” in *Interspeech 2022*, 2022, pp. 2063–2067.
- [13] Y. Higuchi, N. Chen, Y. Fujita, H. Inaguma, T. Komatsu, J. Lee, J. Nozaki, T. Wang, and S. Watanabe, “A comparative study on non-autoregressive modelings for speech-to-text generation,” in *2021 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*. IEEE, 2021, pp. 47–54.
- [14] M. Someki, N. Eng, Y. Higuchi, and S. Watanabe, “Segment-level vectorized beam search based on partially autoregressive inference,” in *2023 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*. IEEE, 2023, pp. 1–8.
- [15] S. Watanabe, T. Hori, S. Kim, J. R. Hershey, and T. Hayashi, “Hybrid ctc/attention architecture for end-to-end speech recognition,” *IEEE Journal of Selected Topics in Signal Processing*, vol. 11, no. 8, pp. 1240–1253, 2017.
- [16] H. Seki, T. Hori, S. Watanabe, N. Moritz, and J. L. Roux, “Vectorized Beam Search for CTC-Attention-Based Speech Recognition,” in *Proc. Interspeech 2019*, 2019, pp. 3825–3829.
- [17] Y. Leviathan, M. Kalman, and Y. Matias, “Fast inference from transformers via speculative decoding,” in *International Conference on Machine Learning*. PMLR, 2023, pp. 19 274–19 286.
- [18] Y. Peng, K. Kim, F. Wu, B. Yan, S. Arora, W. Chen, J. Tang, S. Shon, P. Sridhar, and S. Watanabe, “A Comparative Study on E-Branchformer vs Conformer in Speech Recognition, Translation, and Understanding Tasks,” in *Proc. INTERSPEECH 2023*, 2023, pp. 2208–2212.
- [19] S. Watanabe, T. Hori, S. Karita, T. Hayashi, J. Nishitoba, Y. Unno, N. Enrique Yalta Soplin, J. Heymann, M. Wiesner, N. Chen, A. Renduchintala, and T. Ochiai, “ESPnet: End-to-end speech processing toolkit,” in *Proceedings of Interspeech*, 2018, pp. 2207–2211. [Online]. Available: <http://dx.doi.org/10.21437/Interspeech.2018-1456>
- [20] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, “Librispeech: an asr corpus based on public domain audio books,” in *2015 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, 2015, pp. 5206–5210.
- [21] A. Rousseau, P. Deléglise, Y. Esteve *et al.*, “Enhancing the ted-lium corpus with selected data for language modeling and more ted talks,” in *LREC*, 2014, pp. 3935–3939.
- [22] R. Sonobe, S. Takamichi, and H. Saruwatari, “Jsut corpus: free large-scale japanese speech corpus for end-to-end speech synthesis,” *arXiv preprint arXiv:1711.00354*, 2017.
- [23] H. Bu, J. Du, X. Na, B. Wu, and H. Zheng, “Aishell-1: An open-source mandarin speech corpus and a speech recognition baseline,” in *2017 20th conference of the oriental chapter of the international coordinating committee on speech databases and speech I/O systems and assessment (O-COCOSDA)*. IEEE, 2017, pp. 1–5.